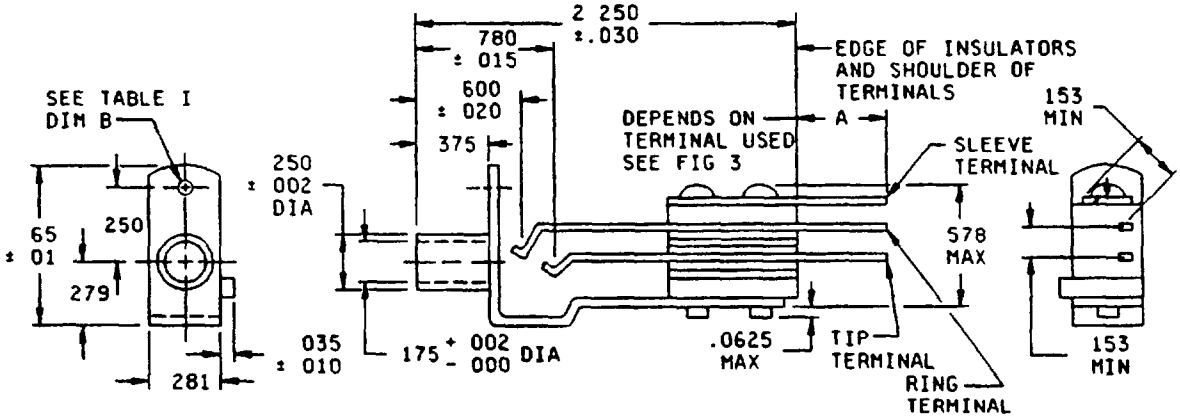
INCH-POUND
MIL-J-641/19E
12 November 1993
SUPERSEDING
MIL-J-641/19D
26 February 1979

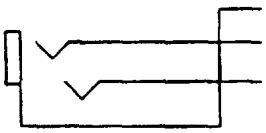
MILITARY SPECIFICATION SHEET

JACKS, (BANTAM) TELEPHONE, 3 CONDUCTOR, SINGLE

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-J-641





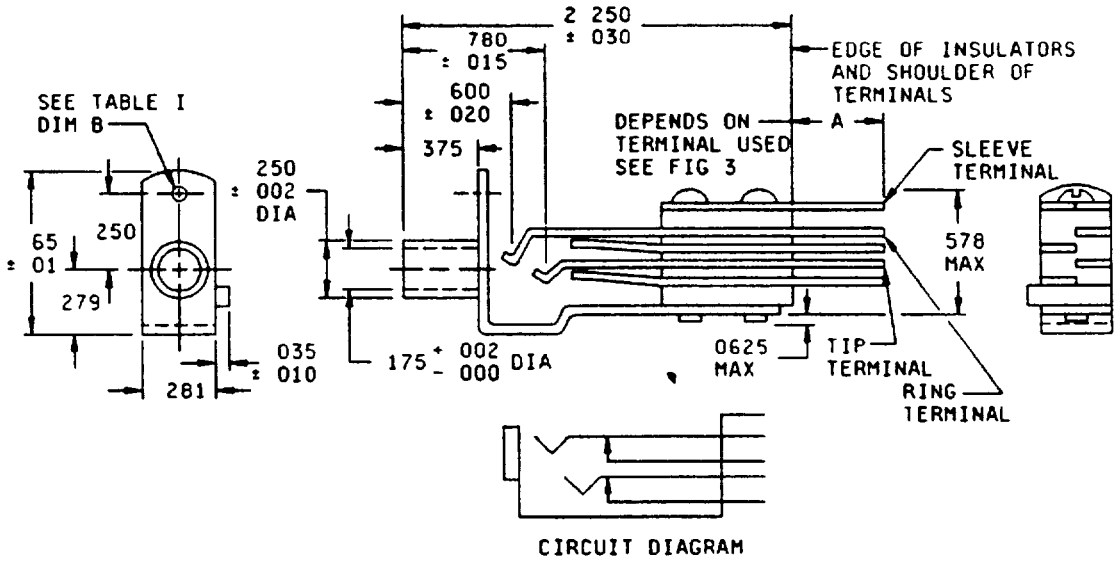
CIRCUIT DIAGRAM

Dash nos -1, -2, -5, and -6 (no breaks)

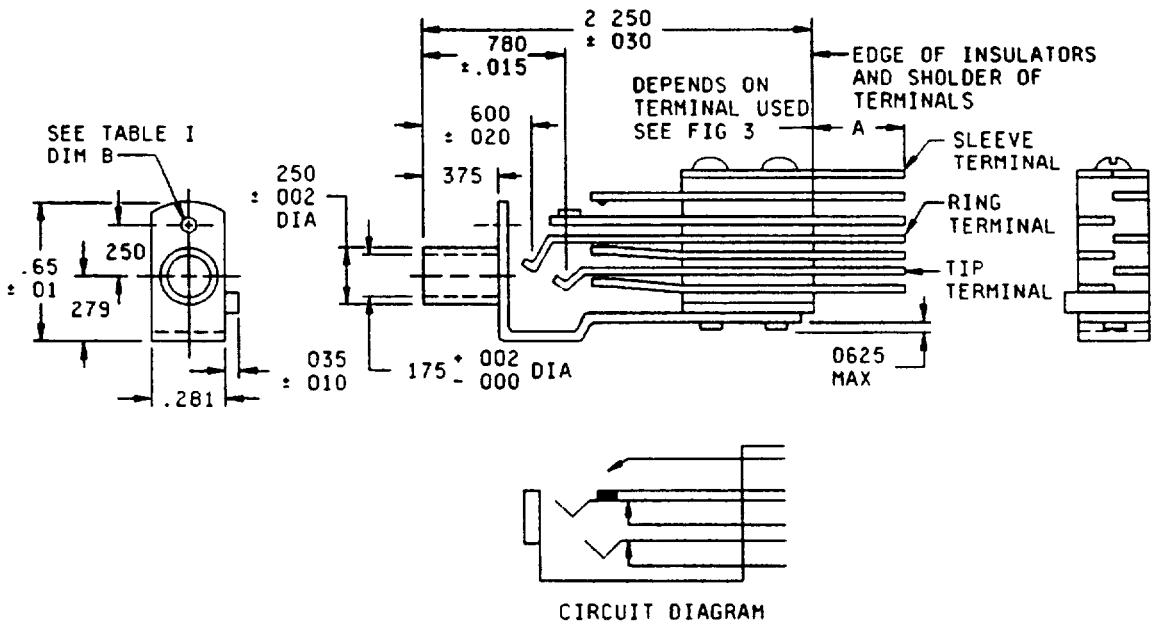
FIGURE 1 Dimensions and configurations

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Dash nos -3, -4, -7, and -8 (2 breaks)



Dash nos -9, -10, -11, and -12 (2 breaks, 1 make)

FIGURE 1. Dimensions and configurations - Continued

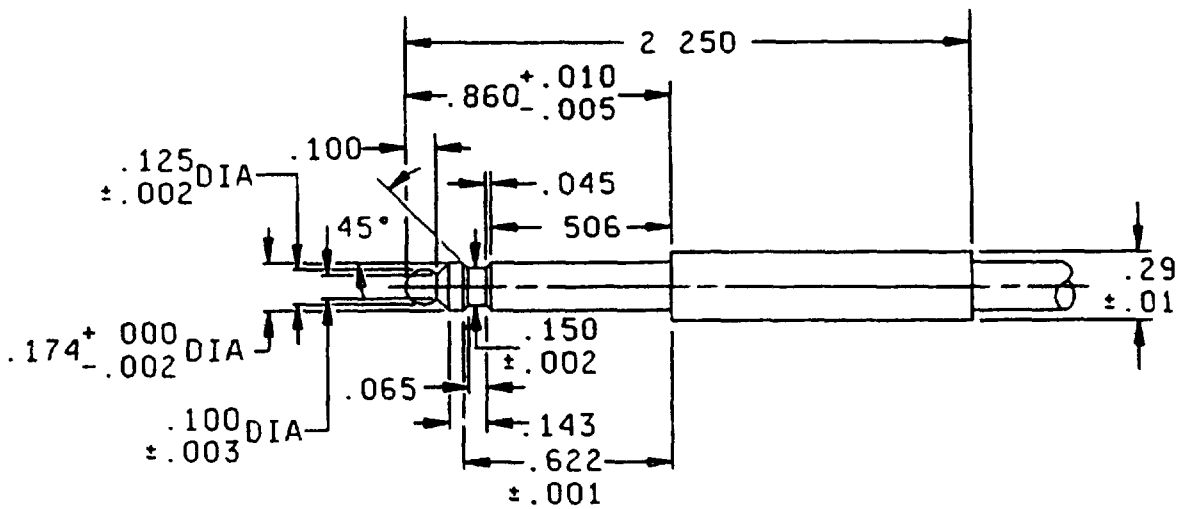
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Inches	mm	Inches	mm
.002	0.05	250	6 35
010	0.25	279	7 09
015	0 38	.281	7 14
020	0.51	375	9 52
030	0 76	578	14 68
.035	0 89	.600	15 24
.0625	1 588	.65	16 5
.153	3 89	.780	19 81
.175	4 44	2 250	57 15

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for information only
3. Unless otherwise specified, tolerance is ± 005 (0 13 mm)
4. All dimensions are after plating
5. All undimensioned pictorial features are for reference purposes only

FIGURE 1 Dimensions and configurations - Continued

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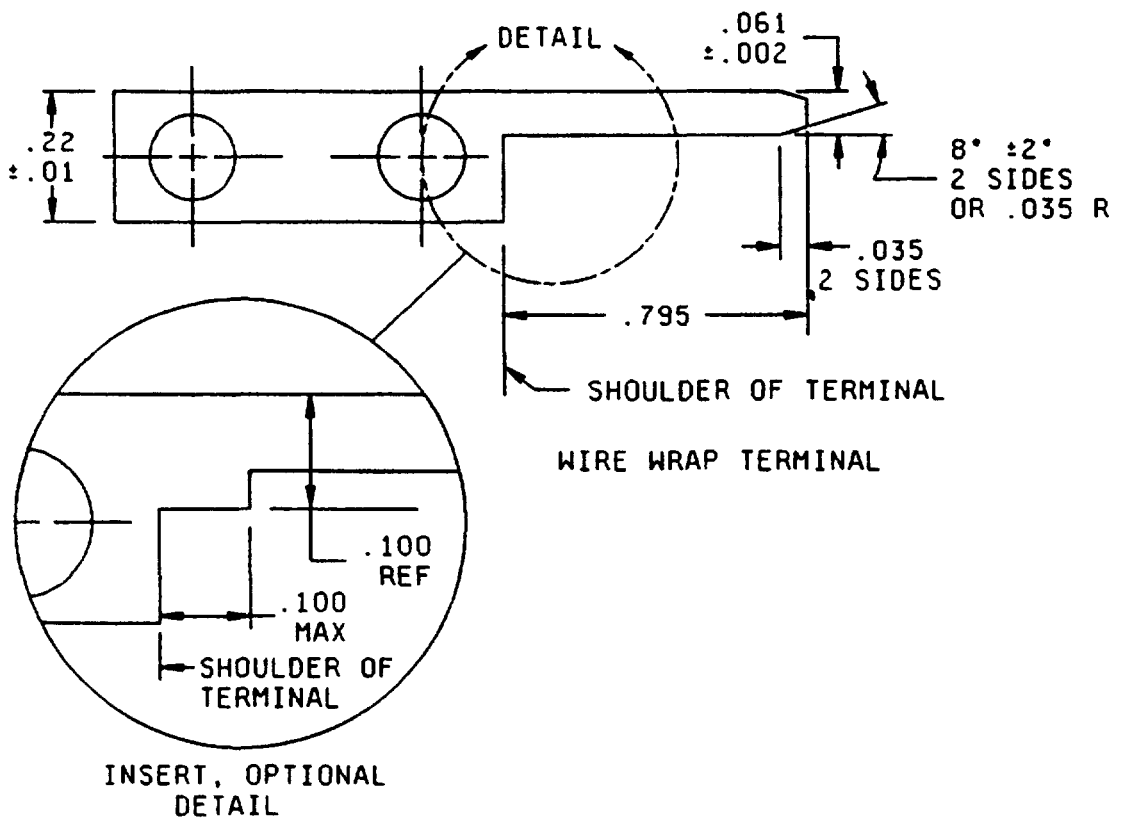


Inches	mm	Inches	mm	Inches	mm
001	0 03	065	1 65	29	7 4
002	0 05	100	2 54	506	12.85
003	0 08	.125	3 18	622	15 80
.005	0 13	143	3 63	860	21 84
010	0 25	.150	3.81	2 250	57.15
.045	1 14	174	4.42		

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for information only
3. Unless otherwise specified, tolerances are ±.005 (0 13 mm) and 1° on angles
4. Finger shall be hardened to Rockwell C 60-62, polished, and then hard-chromium plated to 0003 inch (0 008 mm) maximum thickness
5. All dimensions are after plating
6. Concentricity shall be 003 (0 08 mm) total indicator reading

FIGURE 2. Test plug TP-777

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Inches	mm	Inches	mm
.002	0 05	.100	2 54
01	0 3	22	5.6
035	0 89	795	20 19
.061	1 55		

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for information only
3. Unless otherwise specified, tolerance is ± 005 (0.13 mm)
4. Solder lug terminals (not shown) shall meet the requirements of MIL-J-641, 3 5 5, and the "A" dimension of figure 1 shall be .325 inch (8 26 mm) maximum
5. All dimensions are after plating

FIGURE 3 Terminals

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REQUIREMENTS

Dimensions and configurations See figure 1

Terminals See figure 3 (wirewrap) or MIL-J-641 (solder lug)

Materials
Springs Nickel silver
Contact Palladium or gold alloy

Insertion and withdrawal forces
Insertion force 7 pounds maximum
Withdrawal force: 1.0 pound minimum, 7 pounds maximum.
Test plug TP-777 (see figure 2).

Spring life Test plug TP-777 shall be used

Frame strength. Frame strength test is applicable Load - 7 pounds maximum.

Terminal of tip contact spring shall be on the right side when terminals are viewed from the rear, with the jack in a position such that the contact spring pileup is on top of the jack frame

Cutout in the jack frame shall be positioned so that a minimum of .016-inch clearance exists between the tip of the TP-777 plug and the edge of the frame cutout at all times during insertion of the plug.

Group qualification See table II

All terminals shall be stacked as shown on schematic (see figure 1)

Supersession data See table III

Mating plugs See table IV

Part or Identifying Number (PIN) M641/19 - (dash number from table I).

TABLE I Dash numbers and characteristics

Dash number	Connection	Mounting	Switching action	Terminal	Dimension B
-1	Tip, ring, and sleeve	Single lug	None	Wirewrap	.115 diameter
-2	"	"	None	Solder lug	.115 diameter
-3	"	"	2 breaks	Wirewrap	.115 diameter
-4	"	"	2 breaks	Solder lug	.115 diameter
-5	"	"	None	Wirewrap	.099-48UNC-2B
-6	"	"	None	Solder lug	.099-48UNC-2B
-7	"	"	2 breaks	Wirewrap	.099-48UNC-2B
-8	"	"	2 breaks	Solder lug	.099-48UNC-2B
-9	"	"	2 breaks, 1 make	Wirewrap	.115 diameter
-10	"	"	2 breaks, 1 make	Solder lug	.115 diameter
-11	"	"	2 breaks, 1 make	Wirewrap	.099-48UNC-2B
-12	"	"	2 breaks, 1 make	Solder lug	.099-48UNC-2B

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TABLE II Group qualification

Submission and qualification of the following dash numbers	Qualifies the following dash numbers
-1, -2, -5, or -6	-1, -2, -5, or -6
-3, -4, -7, -8, -9, -10, -11, or -12	-1 through -12

TABLE III. Supersession data

PIN M641/19-	Manufacturer's PIN	Manufacturer's CAGE
5	WTT-32BFM	82389
6	TT-32BFM	82389
7	WTT-34BFM	82389
8	TT-34BFM	82389
9	PJ-805W	70674
10	PJ-805	70674

TABLE IV. Mating plugs

Dash number	Mating plug
-1	M642/13-1, -2
-2	"
-3	"
-4	"
-5	"
-6	"
-7	"
-8	"
-9	"
-10	"
-11	"
-12	"

Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 85

Review activities
Army - AR, AV, ME, MI
Navy - AS, MC
Air Force - 17, 19, 99
DLA - ES

Preparing activity
Army - CR

Agent
DLA - ES

(Project 5935-3910-01)

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